

Introduction:

Lung fibrosis is a persistent scarring of the lung epithelium that worsens over time. It could cause the respiratory structures to become distorted and disordered within the thoracic cavity. Due to the disfiguration and thickening of the lung tissue, the bloodstream is impaired, and the diffusion of carbon dioxide and oxygen exchange is hindered. This leads to intense respiratory failure, causing shortness of breath and limitations in activity. (1) Pulmonary fibrosis may be triggered by previous, unresolved cases of lung inflammation, where disease-causing fibrotic tissue gradually accumulates within the lung over time, weakening the lung's flexibility and strength. However, other forerunners can also trigger fibrosis, including autoimmune diseases and various types of inflammation. (1, 2)

The pulmonary surfactant proteins — SP-A, SP-B, SP-C, and SP-D — play a vital role in the lung's defence mechanism against severe inflammation. These proteins serve two purposes: they maintain alveolar cohesion and regulate immunological responses. Together, these proteins and surfactant lipids form a complex system that lowers surface tension at the lungs' air-liquid interface, preventing alveolar collapse during respiration. (3) In addition to their biophysical functions, surfactant proteins, which bind to pathogens and regulate inflammatory responses, are crucial for innate immunity. Given that these proteins predate the development of vertebrate lungs, their evolutionary history suggests that their primary role may have been in innate protective mechanisms before they were adapted for respiratory functions. (4)

According to a recent study conducted between 2009 and 2020, prevalent and existing cases of idiopathic pulmonary fibrosis have increased due to an ageing population. The lungs remain inflamed due to the disregard of disease in several areas of the world, where interstitial disease continues to aggravate. In countries where antifibrotic treatments are inaccessible or unaffordable, pulmonary disease becomes fatal. However, it remains a relatively rare disease, affected by factors such as age, sex, and other specific conditions. (5)

Recent studies in the literature highlight that surfactant proteins have two sides to their impact on lung health. SP-A inhibits TGF- β 1-driven fibrotic pathways, whereas SP-D reduces fibrosis by regulating macrophage activity. (6) On the other hand, SFTPC and SFTPA2 mutations cause proteostasis to be upset, which leads to endoplasmic reticulum stress and alveolar epithelial cell death, two major causes of familial PF. These results are consistent with biomarker studies that demonstrate a correlation between increasing fibrosis and higher pro-SFTPB levels in serum extracellular vesicles. (6, 7) However, there is a significant knowledge gap in the evolutionary context because the majority of current research focuses on human and murine models. There is currently no thorough investigation comparing surfactant protein sequences across vertebrates to trace evolutionary adaptations against fibrosis, even though SP genes like SFTPB appeared in early vertebrates and that regulatory changes, rather than additional genes, drove lung evolution. (8) Methodological constraints are the cause of this information gap. The majority of phylogenetic studies only examine single taxa rather than combining genomic, proteomic, and functional data, and

traditional multiple sequence alignment techniques have trouble performing deep evolutionary comparisons of hydrophobic proteins like SP-B/SP-C. Therefore, it is still unknown if the antifibrotic qualities of surfactant proteins developed when lungs adapted to terrestrial settings or if contemporary vulnerability is a result of lost ancestral defences. (8, 9) To answer these problems, new methods that combine ancestral protein reconstruction, cross-species sequence alignment, and functional tests are needed. This multifaceted approach is currently not present in the literature.

Overall, this study aims to determine how the evolutionary trajectory of surfactant proteins has influenced their protective roles in lung health and whether it has contributed to reducing vulnerability to lung fibrosis, a disorder characterised by excessive scarring and loss of pulmonary function.

Methodology

Data collection

The nucleotide and amino acid sequences for surfactant proteins A, B, C, and D were collected for four representative vertebrate species: the wild boar (*Sus scrofa*), the house mouse (*Mus musculus*), the brown rat (*Rattus norvegicus*), and humans (*Homo sapiens*). These sequences were retrieved from the NCBI GenBank database by searching for each protein's gene name in combination with the respective species. Only complete coding sequences with verified annotations were included to ensure data quality and comparability. The NCBI Protein database was also used to retrieve protein sequences for each surfactant protein in each species. Lastly, the protein structures were retrieved from both the PDB and AlphaFold databases to obtain the most comprehensive set of structures available online. Regardless of this approach, some species' surfactant proteins remained unfound and were excluded from analysis.

Multiple sequence alignment

To compare the evolutionary conservation and divergence of surfactant proteins among the selected species, multiple sequence alignments were performed using Clustal Omega software. Separate alignments were generated for each protein family (SP-A, SP-B, SP-C, and SP-D), allowing for the identification of conserved and variable regions within each group. Additionally, a combined alignment including all four proteins was conducted to assess broader patterns of conservation across the surfactant protein family. This was achieved by compiling each surfactant protein into a single file to prepare it for the Clustal Omega analysis. Then, on the website itself, the 'RNA' option was selected, along with 'ClustalW with character counts' for the output format. The results view link was then pasted into the Jalview software, where the sequence was colour-coded based on the conserved areas. This process was repeated with the protein sequences as well: the 'Protein' option was selected on the Clustal Omega website to receive data, and the results view link was pasted into the Jalview software, where the sequence was colour-coded by the 'By conservation' setting and the 'Clustal' option. The data was then saved in the form of a PNG.

Phylogenetic tree construction

To facilitate the interpretation of sequence alignments and highlight regions of conservation and variability among the surfactant proteins, the aligned sequences were visualised using Jalview software. Jalview was utilised to generate colour-coded alignment maps, enabling clear identification of the average species distances based on surfactant protein sequences. This was done by selecting all the species on the software, selecting the 'Calculate' option and choosing the 'Calculate Tree, PCA or PaSiMap...' option. In order to produce phylogenetic trees with evolutionary distances, the 'Average Distance' option was selected under the 'Tree' option.

Protein Structure alignment

Following the construction of the phylogenetic trees, the structures of the pulmonary-associated surfactant proteins were obtained from the PDB database. Wild-type structures were identified and downloaded in the Legacy PDB format to ensure that the natural protein in each organism was being studied. These structures were then aligned using the PyMOL application. The solvents were selected and removed to enhance the visualisation. Using the 'Align' function, the SFTPD proteins of each vertebrate in the investigation were compared to the one in *Homo sapiens*. The RMSD score was evaluated for the alignment of each surfactant protein type.

Disease and variant mapping

To identify potential mutations and diseases in the surfactant proteins of *Homo sapiens*, the UniProt database was utilised. After each surfactant protein (SFTPA1-D) was found and its UniProt ID was registered, the 'Disease & Variants' section was located, and the variant IDs were noted. The clinical significance of each variant was also analysed using data about the position in the amino acid sequence where the mutation occurred, allowing for the comparison with the results of the previous tests. A general evaluation of the commonly occurring diseases associated with each variant was made.

Results

MSA of RNA sequences

The multiple sequence alignment of the SFTPA RNA sequences spanned 7,020 nucleotide positions. The alignment of SFTPA RNA sequences for all five species is shown in Figure 1(a). Conserved regions, where all five species aligned, were identified at positions 5,900–6,310 and 4,030–5,710. These conserved segments indicate functional stability across species, suggesting that mutations in these regions could disrupt protein function and contribute to the development of fibrosis. For SFTPB, the MSA covered 14,140 nucleotide positions. The alignment of SFTPB RNA sequences for all five species is shown in Figure 1(b). The longest variable stretches occurred at positions 6,550–14,140 and 1–2,150. Conserved sequences were observed at 2,150–2,390, 2,940–3,020, 3,380–3,510, 4,360–4,540, 4,770–5,060, 5,150–5,260, and 5,800–6,550. The most significant conserved region was

2,220–2,390, representing the longest continuous alignment across all five species. The SFTPC alignment spanned 7,760 positions. The alignment of SFTPC RNA sequences for all five species is shown in Figure 1(c). The longest variable stretch extended from position 1 to 5,520. Conserved sequences were detected at 5,520–5,750, 6,090–6,220, 6,450–6,560, 6,910–7,070, and 7,280–7,500. The most prominent conserved regions were 5,520–5,750 and 6,910–7,070, both of which exhibited the longest continuous alignment. SFTPD required 1,500 positions for alignment. The alignment of SFTPD RNA sequences for all five species is shown in Figure 1(d). A conserved block spanned positions 70–1,370, while the longest variable stretch was 1,370–1,500. The optimal conserved region was 90–1,140, representing the longest uninterrupted alignment. The integrated MSA of all surfactant proteins spanned 15,330 positions. Here, no fully conserved regions were observed. The longest variable stretches were 1–10,050 and 11,890–15,330. Near-conserved segments occurred at 10,050–10,320, 10,440–10,560, and 11,080–11,790.

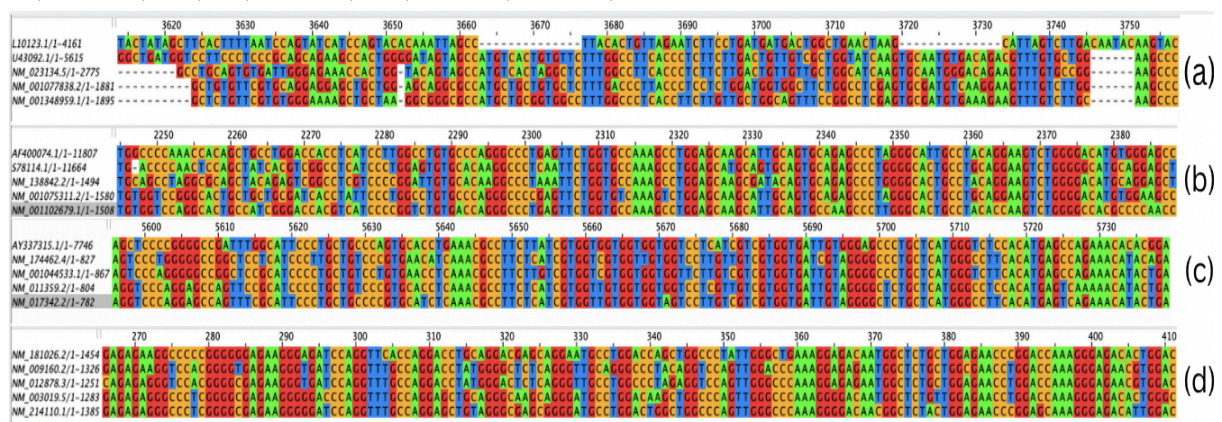


Figure 1. Jalview mRNA sequence alignments. MSAs of the mRNA sequences for the four surfactant proteins (SP-A, SP-B, SP-C, and SP-D) in humans, brown rats, house mice, domestic cattle, and wild boar are shown in this figure. While variable sections are indicated by variable colours, which reflect evolutionary diversity among species, conserved regions are highlighted by uniform colour blocks, which indicate high sequence similarity and functional conservation. SP-A was the least conserved sequence, having many variable regions and few conserved regions with sporadicity. SP-B and SP-C were similar in their level of conservation, being more conserved than SP-A but less conserved than SP-D. SP-D had barely any variable regions.

Hence, through the analysis of the Jalview visualisations, it can be concluded that SFTPD was the most conserved during the evolution of all five vertebrate species (with 87% of the entire sequence conserved), and was most likely evolved via convergent evolution. On the other hand, SFTPC had the least conserved length compared to the whole sequence, at around 11%. Therefore, a mutation in SFTPD would most likely lead to the most severe consequences, possibly resulting in idiopathic pulmonary fibrosis.

MSA of protein sequences

The multiple sequence alignment of the SFTPA protein sequences spanned 260 amino acids. The alignment of SFTPA protein sequences for all five species is shown in Figure 1(a). The

longest variable stretch was from the 1st to the 140th position, after which the sequence remained predominantly conserved across all species. Furthermore, the MSA results for the SFTPb amino acid sequences showed that the longest variable stretch was between the 155th and the 170th positions. The alignment of SFTPb protein sequences for all five species is shown in Figure 1(b). The rest of the 380 amino acid-long alignments remained mainly conserved. The conservation of the SFTPC proteins for all species was also high, as there were no long stretches of variable alignment. The alignment of SFTPC protein sequences for all five species is shown in Figure 1(c). The alignment spanned 190 amino acids. The MSA results for the mRNA sequences of SFTPD remain consistent with those of the protein sequences, as SFTPD remains one of the most highly conserved proteins across all species. The alignment of SFTPD protein sequences for all five species is shown in Figure 1(d). Out of the 390 amino acid long alignment, there are no significant patches of variable alignment. Out of each protein, SFTPD had the highest percentage conservation, while SFTPA had the least percentage conservation since it had the highest area of variable alignments. SFTPC had the highest area of conserved sequences, however.

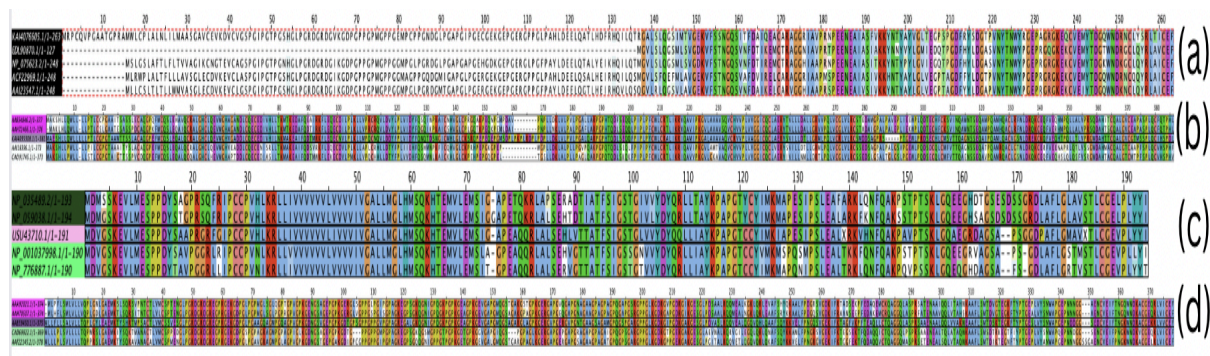


Figure 2. Jalview protein sequence alignments. This figure displays a collection of multiple sequence alignments for each surfactant protein (SP-A, SP-B, SP-C, and SP-D) across all species (human, brown rat, house mouse, domestic cattle, and wild boar), visualised using Jalview software. They are colour-coded according to the conserved areas. The trend for the mRNA sequence alignments is followed here: SP-A is the least evolutionary conserved protein, having predominantly variable regions. The conservation of each protein follows an increasing trend here, where the alignment of the rest of the proteins is mostly conserved, with decreasing variability. SP-D remains one of the most conserved proteins through evolution.

Phylogenetic tree analysis

In the (a) part of Figure 3, for SFTPA, *Bos taurus* [AAI23547.1] and *Sus scrofa* [ACF22969.1] have the most closely related proteins, being the only cluster in the tree. They have a distance of 36.5 from the closest branch point that is not theirs. However, the protein of *Mus musculus* [NP_075623.2] is also related to these two species. *Rattus norvegicus* [EDL90870.1] has the most distantly related protein, having a distance of 455.625 from the closest branch point. The protein most closely related to the one in *Homo sapiens* [KAI4076605.1] is *Mus musculus*, with a distance of only 58.5 from each other. In the (b)

part of Figure 3, for SFTPB, there is one cluster between *Rattus norvegicus* and *Mus musculus*, and the second cluster between *Bos taurus* and *Sus scrofa* remains. *Homo sapiens* has the most distantly related protein from the rest of the species, but it is the closest to the one in *Bos taurus*, with a distance of 108. In the (c) part of Figure 3, for SFTPC, the phylogenetic tree's structure is similar to the one for SFTPB. There are two clusters present: one between *Mus musculus* and *Rattus norvegicus*, and the other between *Bos taurus* and *Sus scrofa*. The protein in *Homo sapiens* remains the most distant from that observed in the other species, also seen in SFTPB. The protein in *Sus scrofa* is the most closely related to the one in *Homo sapiens*, with a distance of 24.75. In the (d) part of Figure 3, for SFTPD, *Sus scrofa* and *Bos taurus* continue to have closely related protein sequences, as they share a cluster. The cluster between *Rattus norvegicus* and *Mus musculus* is also seen in this tree. Hence, *Homo sapiens* remains to have the most distantly related protein. The protein in *Rattus norvegicus* is the most closely related to the one in *Homo sapiens*, with a distance of 159.75.

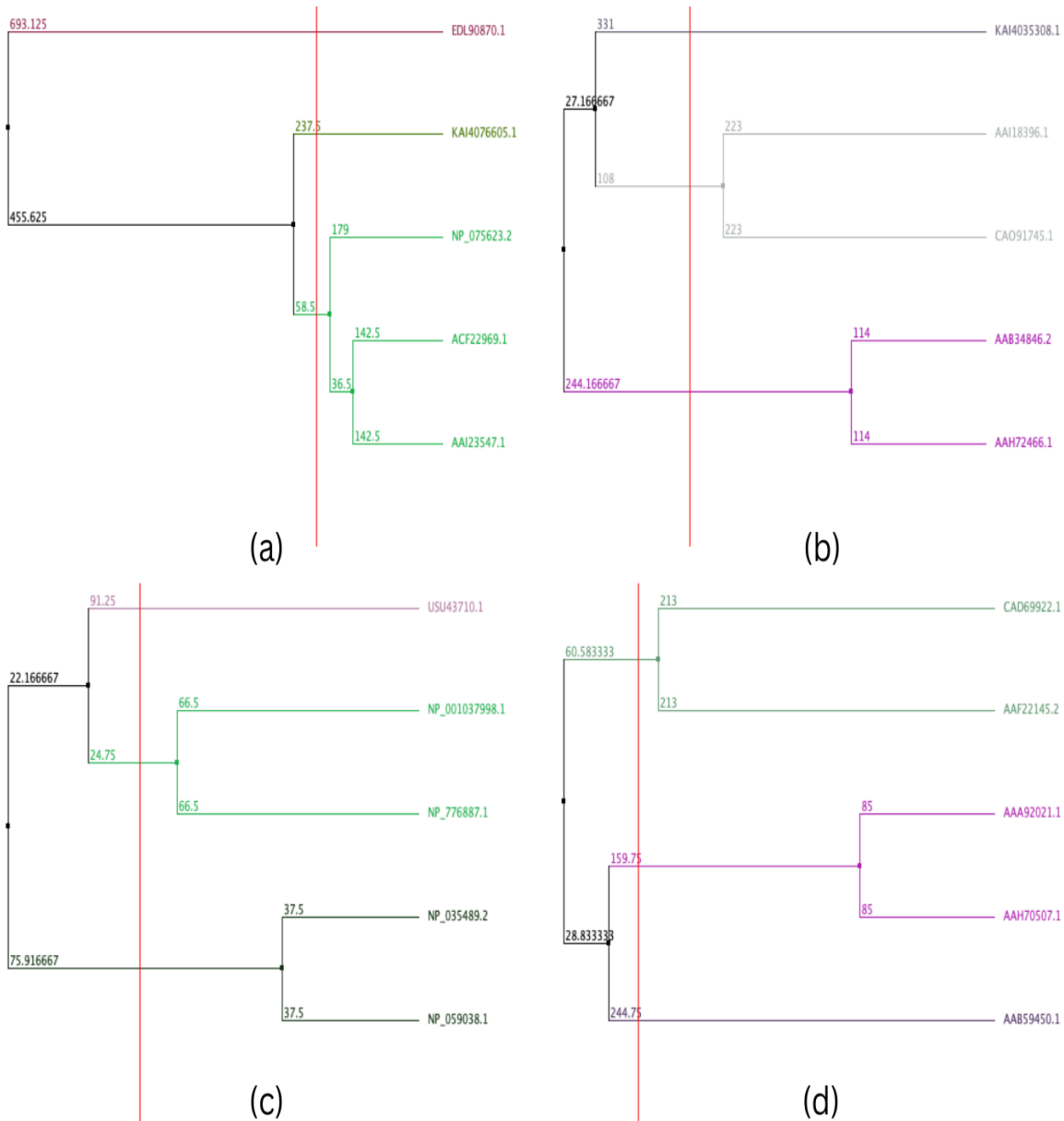


Figure 3. Phylogenetic trees for surfactant proteins visualised in Jalview. This figure shows the phylogenetic trees for each surfactant protein derived from the protein sequence visualisations.

MSA of protein structures

The results of the protein structure alignments were comparable to the multiple sequence alignments of the mRNA and protein sequences. Due to the unavailability of surfactant proteins A, B, and C for all the species, only the alignment of SFTPD was investigated for *Homo sapiens* and *Sus scrofa*. Furthermore, only the A-chain of the SFTPD protein for *Sus scrofa* was found. Regardless, the alignment of these two proteins corresponded to the results on the phylogenetic tree produced on Jalview. The visualisation is seen in Figure 4. The results showed that the SFTPD proteins in *Homo sapiens* and *Sus scrofa* were distantly related, also seen in Figure 3 (d). This was confirmed by the RMSD (root mean square deviation) value of 0.538 Å (ångström). Since an RMSD value from 0-2 Å is considered highly conserved, 2-4 Å is considered moderately similar, and greater than 4 Å means significant structural differences, it was sufficient to conclude that the SFTPD proteins in the selected species were distantly related.

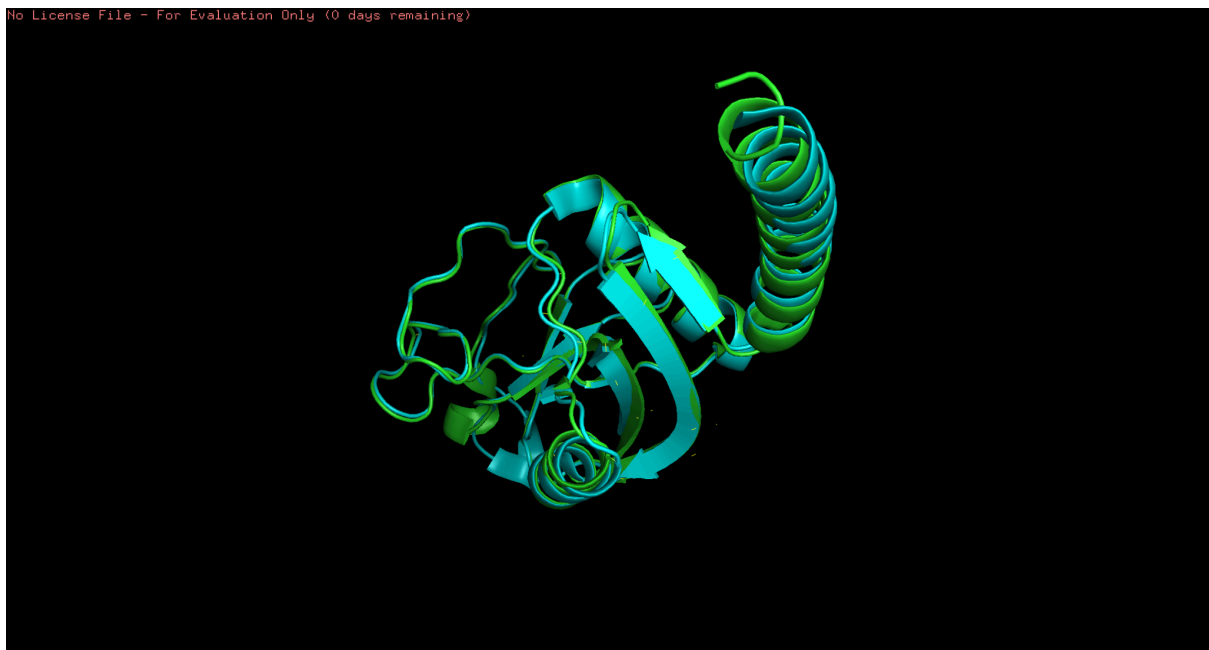


Figure 4. PyMOL alignment of SFTPD structures. This figure shows the aligned A-chains of the surfactant protein D of *Homo sapiens* and *Sus scrofa*. The RMSD score of 0.538 is seen in the image. The blue A-chain represents the surfactant protein D in wild boars, and the green A-chain represents the surfactant protein D in humans. Both structures are seen to have alpha helices, beta folded sheets, and triple-helical structures.

Analysis of disease and variant mapping

For the SFTPA1 protein in *Homo sapiens*, the variants as provided by UniProt include VAR_086118, VAR_086119, VAR_086120, and VAR_086121. These occurred in positions

178, 208, 211, and 225 in the amino acid sequences, respectively. For SFTPA1, data on the diseases and their clinical significance were not mentioned on the UniProt website. Hence, conclusions could not be made on the benignity or pathogenicity of the variants. The variants for the SFTPA2 protein were VAR_086122, VAR_086123, VAR_086124, VAR_063519, VAR_063520, VAR_086125, VAR_086126, VAR_086127, VAR_086128, and VAR_086129. The positions of these mutations were 171, 178, 181, 198, 231, 233 [thrice], 238, and 242, respectively. The pathogenic diseases associated with these mutations were interstitial lung disease and pulmonary fibrosis, indicating that SFTPA2 is more susceptible to lung conditions. Furthermore, SFTPB had all natural variants, causing most of them to be benign. The variants included VAR_006948, VAR_013099, VAR_006950, VAR_006949, VAR_036856, and VAR_013100. These mutations were present in the 131st, 176th, 228th [twice], 236th, and 272nd positions, respectively. Most of the diseases were linked to some version of surfactant metabolism dysfunctions and alveolar malfunctions, which were predominantly benign but also included likely pathogenic and fully pathogenic variants. The opposing trend was observed in SFTPC, where most mutations were pathogenic to pulmonary factors. However, the types of disease were similar in SFTPB and SFTPC. The latter was associated with pulmonary fibrosis, unlike in SFTPB. The variant IDs were VAR_036855, VAR_026753, VAR_026754, VAR_026755, and VAR_026756. Their positions in the protein's amino acid sequence were 66, 73, 116, 167, and 188, respectively. Lastly, similar to SFTPB, the variants for SFTPD were all naturally occurring ones. They include VAR_020937, VAR_020938, VAR_020939, VAR_020940, and VAR_020941. In the amino acid sequence, they occurred at positions 31, 123, 180, 290, and 309, respectively. While all the disease names were unspecified, the clinical significance of all variants was noted as benign. This data further validates the hypothesis that SFTPD is the most vital pulmonary-associated surfactant protein and has evolved to adapt against mutations.

The evaluation of the disease and variant mapping of the surfactant proteins in *Homo sapiens* concluded that there was high evolutionary conservation of SFTPD. The lack of pathogenic variants as seen on UniProt indicates surfactant protein D's strong purifying selection, implying that damaging mutations were not tolerated in the conserved regions through the course of evolution. Contrastingly, surfactant proteins A, B, and C allow mutations associated with pulmonary diseases as their sequences were amongst the least conserved across the species chosen for the investigation.